ALEXANDRE AMICE

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EDUCATION

Massachusetts Institute of Technology (M.I.T) August 2020 - May 2025 (expected)
Doctor of Philosophy: Electrical Engineering and Computer Science
Advisors: Pablo Parrilo and Russ Tedrake

University of Pennsylvania August 2018 - May 2020
Masters of Science in Engineering: Robotics
Thesis: Optimal Constraint Relaxation: Theory and Application to Problems in Robotics
Advisors: Alejandro Ribeiro and C.J. Taylor

University of Pennsylvania August 2016 - May 2020
Bachelors of Science in Engineering: Electrical Engineering and Mathematics *Summa Cum Laude*

RESEARCH INTERESTS

My research focuses on fundamental advances in optimization and computational algebra to push the frontiers of optimal and efficient decision-making in a broad range of engineering fields. I am particularly motivated by problems in the control and verification of complex dynamical systems arising in robotics.

ACADEMIC PUBLICATIONS

(* denotes co-first authorship/equal contribution)

Journals

[1] Hongkai Dai, **Alexandre Amice***, Peter Werner, Annan Zhang, and Russ Tedrake. “Certified polyhedral decompositions of collision-free configuration space”. In: *The International Journal of Robotics Research* ***Invited Paper*** (2023).

[2] Lujie Yang, Hongkai Dai, **Alexandre Amice**, and Russ Tedrake. “Approximate Optimal Controller Synthesis for Cart-Poles and Quadrotors via Sums-of-Squares”. In: *IEEE Robotics and Automation Letters* (2023).

[3] Luiz FO Chamon, **Alexandre Amice**, and Alejandro Ribeiro. “Approximately supermodular scheduling subject to matroid constraints”. In: *IEEE Transactions on Automatic Control* 67.3 (2021).

Conferences

[1] **Alexandre Amice**, Peter Werner, and Russ Tedrake. “Certifying Bimanual RRT Motion Plans in a Second”. In: *International Conference on Robotics and Automation* (2024).

[2] Bernhard P Graesdal, Shao YC Chia, Tobia Marcucci, Savva Morozov, **Alexandre Amice**, Pablo A Parrilo, and Russ Tedrake. “Towards Tight Convex Relaxations for Contact-Rich Manipulation”. In: *Robotics: Science and Systems* (2024).

[3] Savva Morozov, Tobia Marcucci, **Alexandre Amice**, Bernhard P Graesdal, Rohan Bosworth, Pablo A Parrilo, and Russ Tedrake. “Multi-Query Shortest-Path Problem in Graphs of Convex Sets”. In: *Workshop on the Algorithmic Foundations of Robotics (WAFR)* (2024).

[4] Peter Werner, **Alexandre Amice**, Tobia Marcucci, Russ Tedrake, and Daniela Rus. “Approximating Robot Configuration Spaces with few Convex Sets using Clique Covers of Visibility Graphs”. In: *International Conference on Robotics and Automation* (2024).

[5] **Alexandre Amice** and Pablo Parrilo. “Solving Least Squares Problems on Partially Ordered Sets”. In: *IEEE 60th Conference on Decision and Control (CDC)*. IEEE. 2022.

[6] **Alexandre Amice***, Hongkai Dai*, Peter Werner, Annan Zhang, and Russ Tedrake. “Finding and Optimizing Certified, Collision-Free Regions in Configuration Space for Robot Manipulators”. In: *Workshop on the Algorithmic Foundation of Robotics (WAFR)* ***Outstanding Paper Award***. 2022.

[7] Lujie Yang, Kaiqing Zhang, **Alexandre Amice**, Yunzhu Li, and Russ Tedrake. “Discrete Approximate Information States in Partially Observable Environments”. In: *2022 American Control Conference (ACC)*. IEEE. 2022, pp. 1406–1413.

[8] Luiz FO Chamon, **Alexandre Amice**, Santiago Paternain, and Alejandro Ribeiro. “Resilient control: Compromising to adapt”. In: *59th IEEE Conference on Decision and Control (CDC)* ***Roberto Tempe Award Nominee***. IEEE. 2020.

[9] Luiz FO Chamon, **Alexandre Amice**, and Alejandro Ribeiro. “Matroid-constrained approximately supermodular optimization for near-optimal actuator scheduling”. In: *IEEE 58th Conference on Decision and Control (CDC)*. IEEE. 2019.

Preprints

[1] Savva Morozov, Tobia Marcucci, Bernhard Paus Graesdal, **Alexandre Amice**, Pablo A Parrilo, and Russ Tedrake. “Mixed discrete and continuous planning using shortest walks in graphs of convex sets”. In: *arXiv preprint arXiv:2507.10878* (2025).

EXPERIENCE

MIT CSAIL and LIDS: Research Assistant August 2020 - Present

- Designed algorithms for succinctly decomposing robot configuration spaces into convex sets. Leveraged these descriptions to generate advances in motion planning algorithms for contact-rich systems.
- Contributed to Sums-of-Squares (SOS)/Semidefinite programming (SDP) formulations of motion planning through contact, optimal feedback control, and provably collision-free motion planning algorithms. Implemented formulations directly in C++ to enable realistic deployment on systems.
- Research on the exploitation of structure to accelerate solving mathematical program arising in the context of control theory and robotic applications.
- Provided mentorship to four junior graduate students and directly supervised research work of two undergraduate students.

Boston Dynamics AI Institute: Research Internship June 2023-August 2023

- Member of the Ultra-Mobility Vehicle team advised by Gabe Nelson and Farbod Farshidian.
- Research in the fundamental tradeoffs in the design of tracking controllers for non-linear systems with rapidly evolving dynamics.

Alelab University of Pennsylvania: Research Assistant Jan 2018 - July 2020

- Research in discrete and convex optimization with applications in optimal control, multi-agent control, machine learning, and computer vision.
- Completed master’s thesis on optimal trade-offs in overspecified robotics problems. Applied novel theory to credibility assignment problems in machine learning and resilient control in teams of drones.

Uber Advanced Technologies Group: Robotic Perception Intern May 2019 - August 2019

- Designed and implemented a novel radar tracking algorithm in Python enabling onboard vehicle radars to perform instance segmentation. Decreased the time from first detection to full-state estimation of objects in the environment by a factor of six.
- Refactored C++ based radar tracking pipeline to allow sensor fusion across radars. Improved performance on proprietary metrics by 20% while reducing computational overhead.

SOFTWARE

Drake

- Core development member of Drake's (an open source model-based simulator for robotics) optimization and geometry toolboxes.
- Contributed extensively to the optimization parser, symbolic algebra, symbolic kinematics, geometry modelling, and planning toolkits in both C++ and Python.

SERVICE AND LEADERSHIP

Conference Organizing

- Neurips 2025: ScaleOPT Workshop Organizing Committee

Conference Reviewing

- IEEE International Conference on Robotics and Automation (ICRA) 2023, 2024
- IEEE Robotics and Automation Letters (RA-L) 2023
- IEEE Transaction on Robotics (T-RO) 2023, 2022
- IFAC Automatica 2023
- International Conference on Machine Learning (ICML) 2023
- Journal of Numerical Algebra, Control, and Optimization 2024
- IEEE Conference on Decision and Control (CDC) 2022

Graduate Application Assistance Program: Mentor — August 2020 - 2023

- MIT: Worked with eight prospective graduate students from underrepresented backgrounds in STEM to strengthen their graduate and fellowship application packages.
- University of Pennsylvania: Launching a graduate application assistance program at the University of Pennsylvania to match prospective graduate students with young alumni to provide mentorship throughout the graduate application process.

NCAA Varsity Athletics: Men's Foil — August 2016 - May 2020

- Dedicated fourteen hours per week to team practice, strength training, and individual coaching sessions. Improved team dynamics through social outings and assisted underclassmen with scheduling and courses.
- Voted most dedicated fencer of 2016-2017 by captains and coaches for outstanding efforts and contributions to the team.

TECHNICAL STRENGTHS

Languages	English (Native Speaker), French (Native Speaker), Spanish (Proficient)
Computer Languages	Python, C++, Julia, MATLAB, Java
Libraries	Drake, CVXPY, PyTorch

TEACHING

Applied Convex Optimization: Lecturer — IAP 2022

MIT 6.S098

- Designed and taught month long course on the fundamentals of convex optimization. Demonstrated applications in finance, control theory, civil planning, and machine learning.
- Overall course quality was rated 6.0/7.0 and overall lecture quality was rated 6.6/7.0 by the class of 10 students.

Algebraic Techniques And Semidefinite Optimization: Teaching Assistant — Spring 2023

MIT 6.7230 — *Supervisor: Pablo Parrilo*

- Supported a class of 25 students for advanced graduate course about the interplay of algebra, algebraic geometry, and convex optimization.
- Overall instruction quality was rated 6.7/7.0.

Underactuated Robotics: Teaching Assistant — Spring 2022
MIT 6.832 — *Supervisor: Russ Tedrake*

- Supported a class of 90 students through office hours for graduate qualifying course in underactuated dynamics and control. Taught review sessions in prerequisite optimization theory to improve student outcomes.
- Overall instruction quality was rated 6.2/7.0. Commended by multiple students for dedication to the course in written reviews.

Modern Convex Optimization: Teaching Assistant — Spring 2020
University of Pennsylvania ESE 605 — *Supervisor: Nikolai Matni*

- Assisted students through office hours in a second year graduate course in convex optimization theory.
- Taught review sessions on prerequisite mathematical foundations to improve student retention.
- Produced solution sets and graded homeworks and exams for a course of 53 students.

Control of Autonomous Systems: Teaching Assistant — Fall 2019
University of Pennsylvania ESE 421 — *Supervisor: Bruce Kothmann*

- Assisted in designing undergraduate labs teaching the basics of robotic perception and control theory. Attended four hours of lab a week to review essential material, lead code design session, and assist students in their development of a rudimentary autonomous vehicle.
- Hosted office hours for four hours a week to provide individual feedback. Taught coding workshops to familiarize students from mechanical, electrical, and computer engineering with the basics of coding in Arduino, C++, and Python. Introduced students to common libraries and modern coding standards.

THESES

Optimal Constraint Relaxation: Theory and Application to Problems in Robotics — 2019-2020
Masters Thesis — *Advisors: Alejandro Ribeiro and C.J. Taylor*

- Proved a novel connection between the optimal dual variables and the optimal slacks of a soft-constrained optimization problem and used this connection to provide a novel algorithm for solving the optimal constraint relaxation problem.
- Provided novel definition of resiliency in control systems based on the relaxing of problem constraints. Applied the theory of resiliency to multi-drone formation simulations using ROS, Gazebo, and Python demonstrating the resilience property's ability to automatically design novel maneuvers for the formation.
- Reformulated the classic machine learning problem of the joint minimization of loss and model risk as the constrained minimization of model risk subject to achieving a specified loss on each training example. Implemented a novel optimizer in PyTorch for training models using this formulation.
- Applied the theory of optimal constraint relaxation to the classification of the Fashion-MNIST dataset and demonstrated the novel formulation's ability to detect mislabelled data, improving classification robustness and prediction confidence.